

Travelers of the World

EVERY YEAR, NEARLY HALF of all the birds in the world set off on journeys, called migration, to find food and water. Migration usually takes place at night, but some birds, such as swallows, migrate during the day. Migration journeys are often very dangerous for

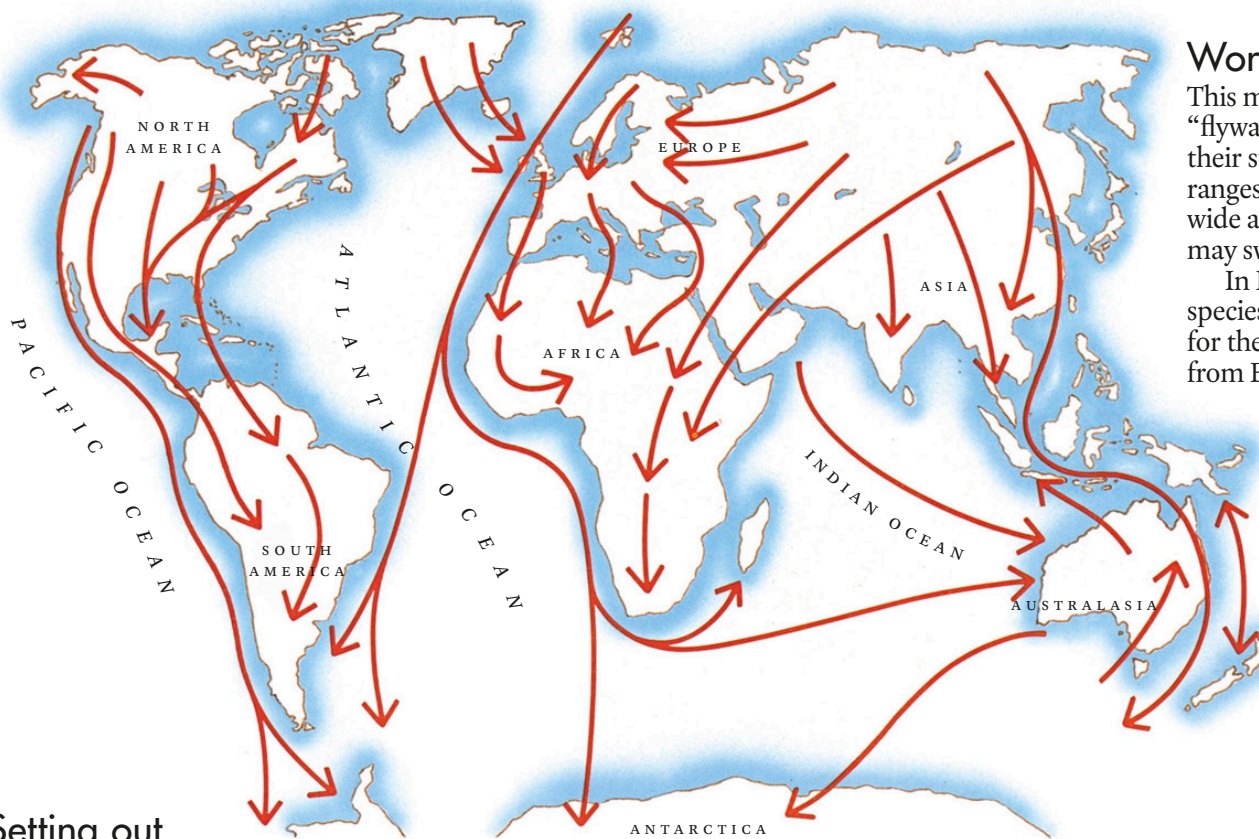
birds, and millions never reach their destinations. They may run out of energy or be unable to find enough food to keep going. Many migrant birds are also killed by bad weather or by people and other predators lying in wait along traditional routes.

World migration routes

This map shows the main migration routes or “flyways” that birds follow when moving from their summer breeding grounds to their winter ranges (although birds generally move over a wide area rather than a narrow path and they may switch from one flyway to another).

In North America, two-thirds of all the bird species that spend the summer there travel south for the winter. Another major winter route is from Europe to Africa. Many birds prefer to avoid crossing the Mediterranean Sea because there are no rising air currents to help them soar and glide along. So they cross it at its narrowest points—for example by the Strait of Gibraltar.

The center of huge continents, such as Asia, are good places for birds to feed and nest in spring and summer, but they get very cold in winter. So many birds migrate from the heart of Asia to the coasts.



Setting out

Before migration, birds eat as much as they can to build up fat reserves in their bodies and give them enough energy to keep going. They usually grow new feathers for the journey, too. Birds know when to migrate because of changes in the weather and also because of a biological “clock” in their brain which responds to changes in daylight hours. As the days get shorter in the fall, birds such as these barn swallows become restless, and get ready to set off by gathering in flocks.

As they migrate, the swallows feed on flying insects to keep up their strength.

Barn swallow
(*Hirundo rustica*)

Finding the way

There is still a lot we do not understand about how birds find their way, or navigate, during migration. They seem to know instinctively which way to go because many young birds make migration journeys for the first time without the help of adults.

Birds also steer by the sun, stars, and the Earth’s magnetic field. A few birds, such as shearwaters and petrels, pick up scents carried by the wind and use these to navigate their way across open oceans. Birds that migrate by day probably find their way year after year by following familiar landmarks, such as river valleys, mountain ranges, and coastlines.

Swallows have long, pointed wings for fast and powerful flight. It takes them five or six weeks to fly from Europe to Africa.